Single cells: the matrix, the clusters, and the statistic that breaks

- Tek hücreler: matris, kümeler ve bozulan istatistik
- BIB Week 10

Ad of the Week #10

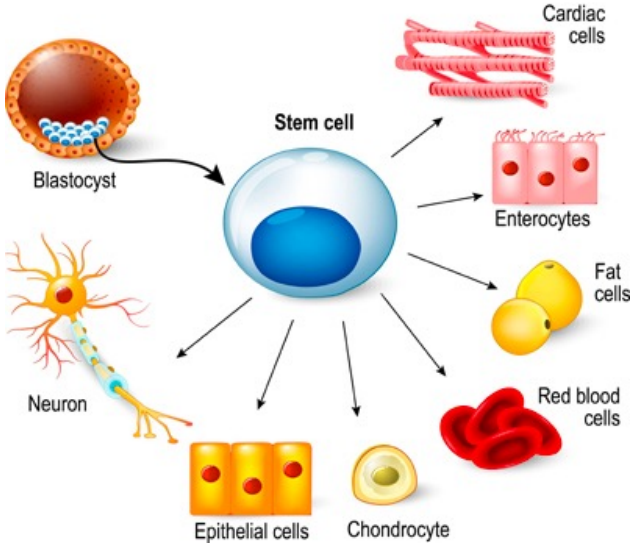
- Skills Ledger: tick single-cell analysis and experimental design

One genome, many cell types

- Nearly every cell in your body carries the same genome
- A neuron and an enterocyte differ in which genes they express, and how much
- That combination — the transcriptome (transkriptom) — is each cell's signature
- One fertilised egg becomes every tissue; the story is written one cell at a time



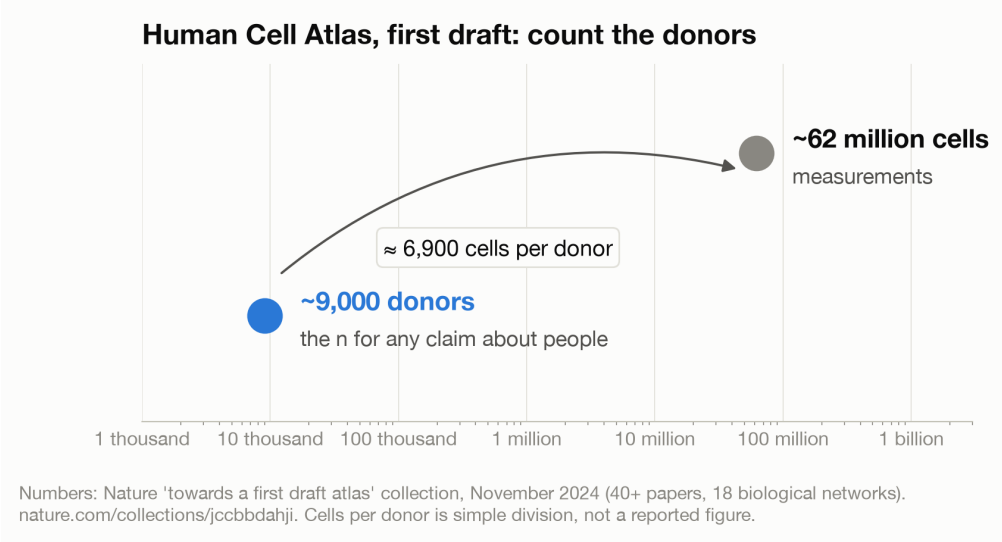
One cell to begin with (2025 deck, slide 2)



One genome, many phenotypes (2025 deck, slide 3; image: bio-rad-antibodies.com)

Atlases in 2026: count the donors, not the cells

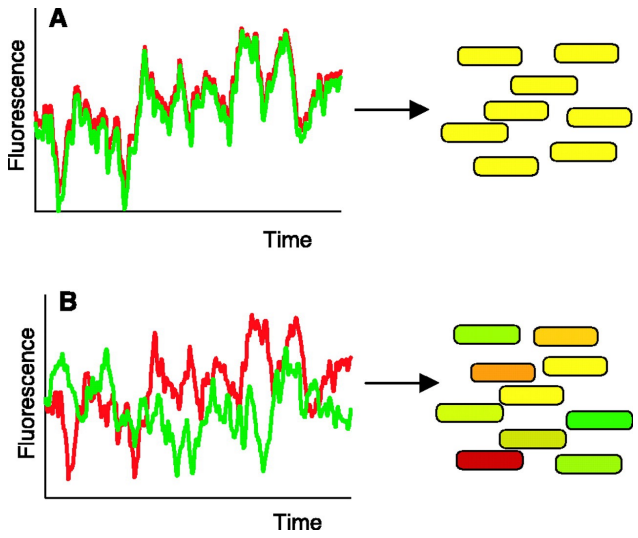
- GTEx, Genotype-Tissue Expression: bulk RNA-seq across tissues and donors
- FANTOM, Functional Annotation of the Mammalian Genome: where transcription starts
- Human Cell Atlas draft (Nature, Nov 2024): ~62 M cells, ~9,000 donors, 18 networks
- Tabula Sapiens 2.0 (Cell, 2 Sept 2026): a multi-organ human reference
- Millions of cells, thousands of donors: for a claim about people, which is n?



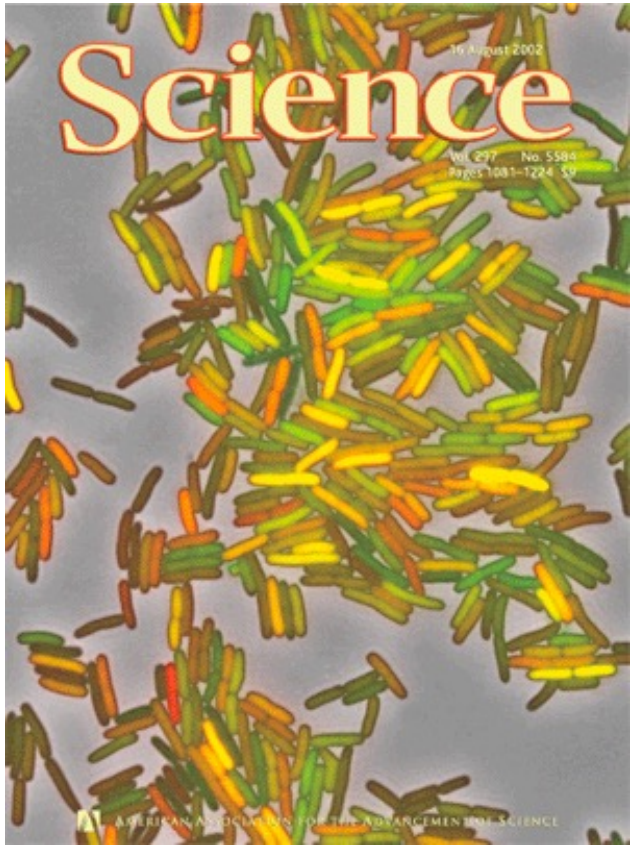
HCA first draft: ~62 million cells, ~9,000 donors (Nature collection, Nov 2024)

Why one cell at a time: noise in identical bacteria

- Two copies of one promoter, two colours, in genetically identical E. coli
- Deterministic expression would make every cell equally both colours: yellow
- They are not: same genome, different amounts — intrinsic noise (iç gürültü)
- A bulk measurement reports the average and calls it "the cell"
- Elowitz et al., Science 2002



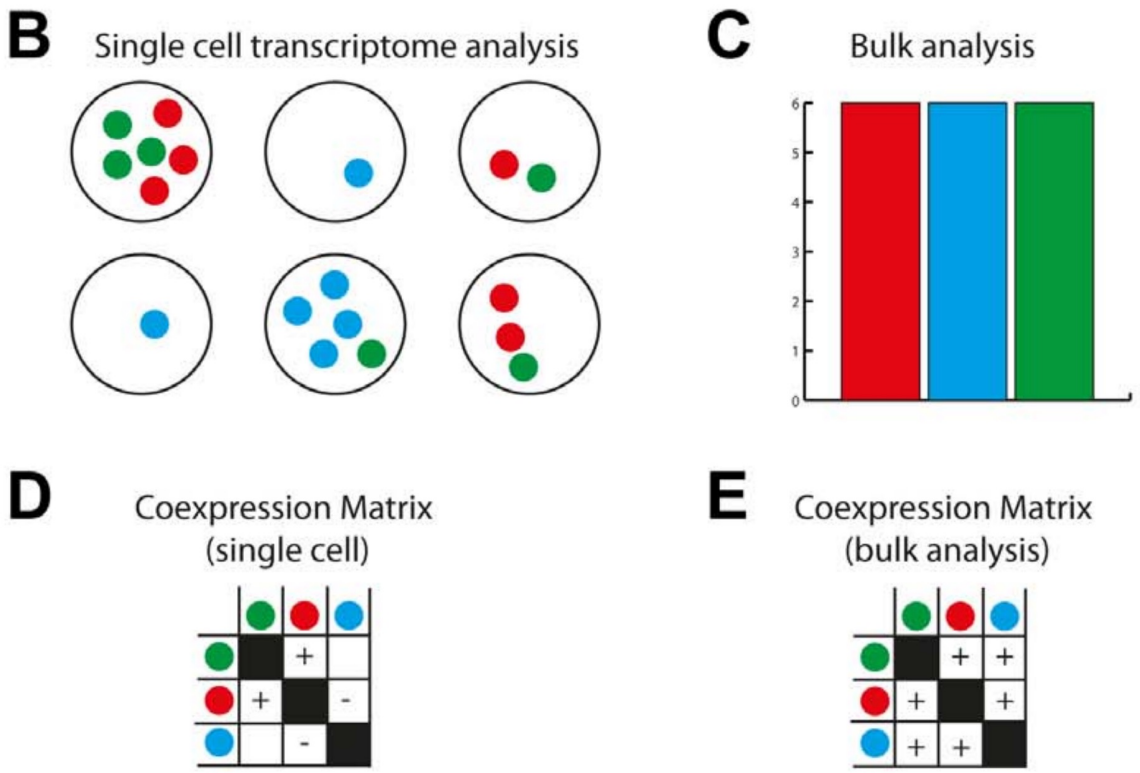
Panel A: the two reporters move together, every cell yellow; panel B: they do not (Elowitz et al. 2002)



Science, 16 August 2002: the cover

What a bulk average hides

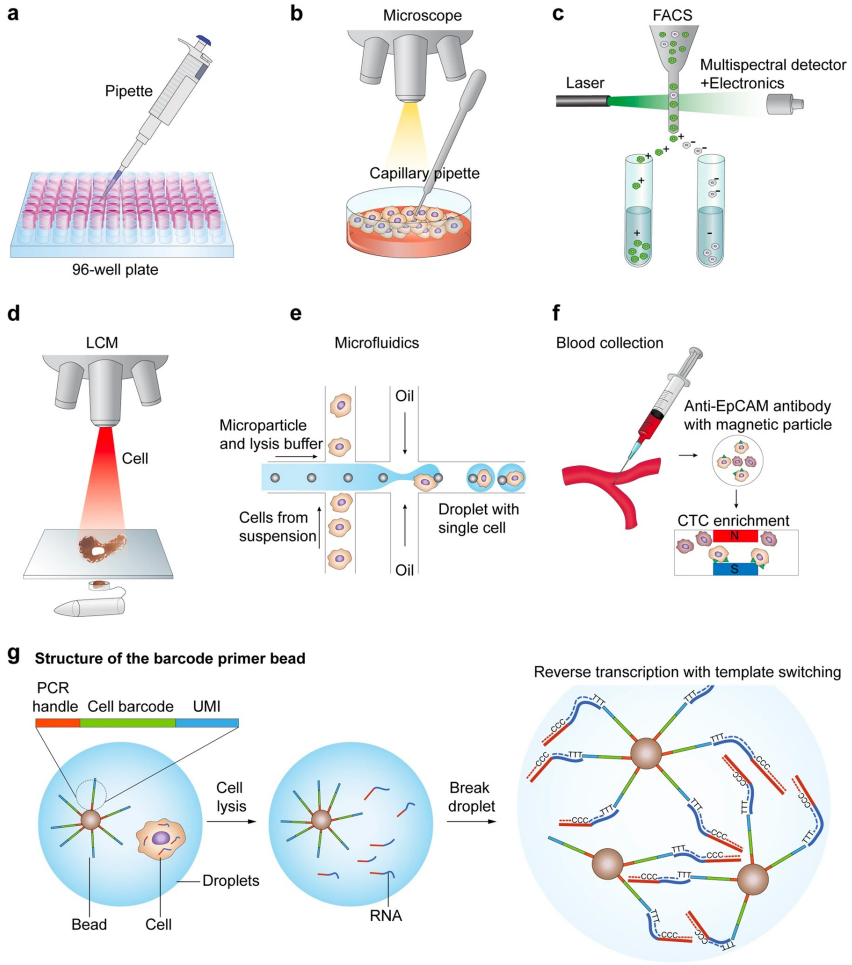
- Six cells, three genes: A and B co-occur; B and C exclude each other
- Bulk sees equal totals and calls every pair positively co-expressed
- Averaging can reverse a relationship, not only blur it
- Resolution decides which conclusion you are allowed to draw
- Macaulay & Voet, PLoS Genet 2014



Single cell vs bulk: the co-expression sign flips (Macaulay & Voet 2014; green A, red B, blue C)

HISTORY: getting one cell into one reaction

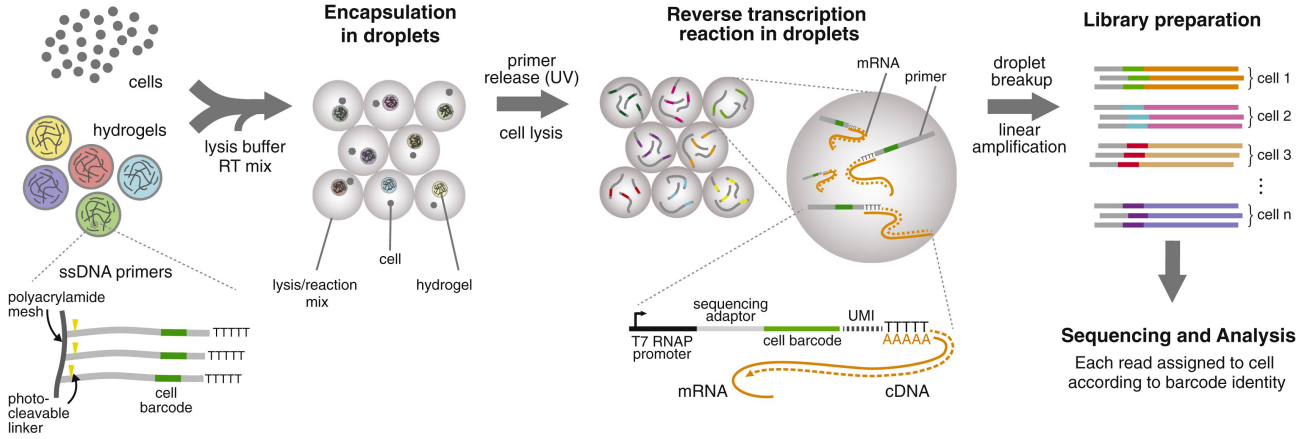
- By hand or sorter: dilution, micromanipulation, FACS, laser capture (LCM)
- One cell per well: hundreds of cells, not hundreds of thousands
- Microfluidics: one cell per droplet, thousands of cells per run
- Panel g survived: a bead with a cell barcode (hücre barkodu) and a UMI
- Hwang et al., Exp Mol Med 2018



HISTORY — collection methods (a–f) and the barcoded bead (g) (Hwang et al., Exp Mol Med 2018)

HISTORY: droplet barcoding (inDrop, 2015)

- Each droplet: one cell + one bead; all its primers share one barcode
- Every cDNA made there inherits the barcode; droplets are then pooled
- Reads are sorted back to cells by barcode; UMIs undo PCR copies
- The principle is right: every droplet platform still uses it
- The products are gone: inDrop is no longer sold



HISTORY — inDrop droplet barcoding (Klein et al., Cell 2015)

Choosing a platform in 2026: four questions

- 1. Live or fixed? Live-cell kits are out for fixed tissue; FFPE needs probes
- 2. Cells per run, cost per cell — set by indexing and probes
- 3. Sensitivity vs doublets (çift hücre)
- 4. What rides along? Receptors, protein, chromatin: per kit
- 10x acquired Scale (Aug 2025); Illumina acquired Fluent (Jul 2024)

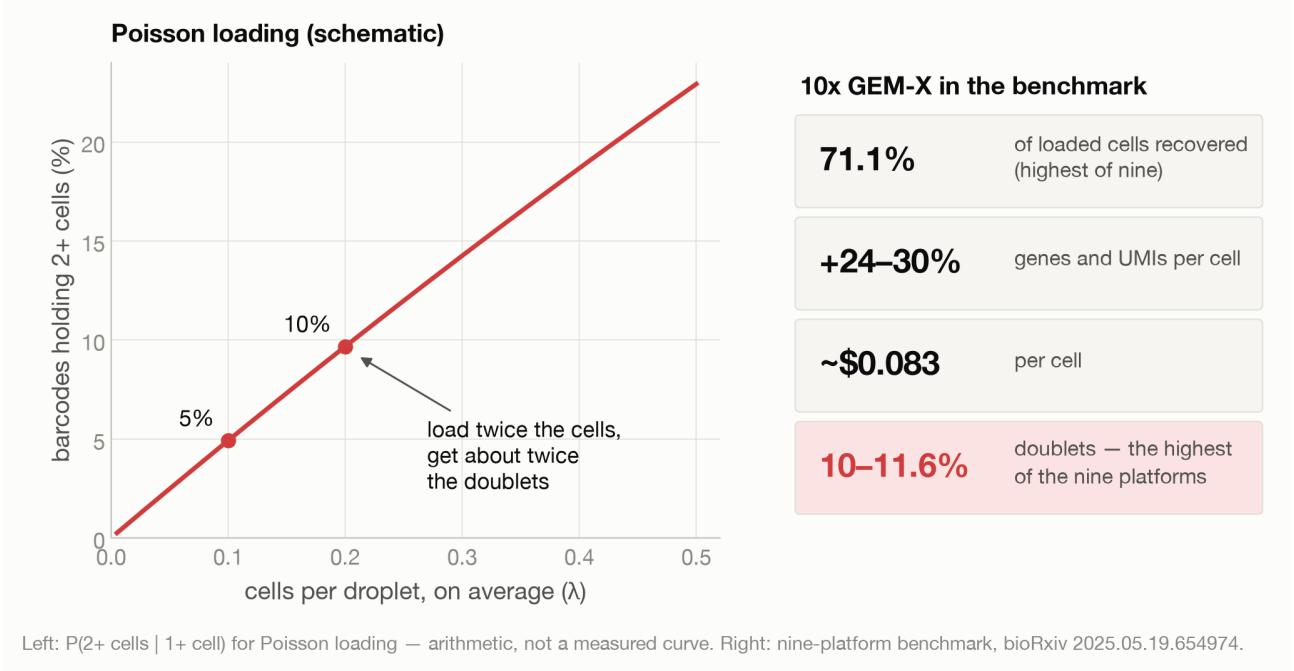
platform	1 live or fixed?	2 per run · per cell	3 sensitivity vs doublets	the catch
10x GEM-X	live cells, droplets (fresh or viably frozen)	~$0.083 per cell 71.1% of loaded cells kept	+24–30% genes/UMIs doublets 10–11.6%	highest doublet rate: you pay in QC
10x Flex	fixed or FFPE, probes samples can be banked	16-plex, ~2.56 M/run $0.107 per cell	88.9% of reads used	dearest per cell; only what is probed
Parse Evercode WT PENTA v3	fixed cells or nuclei no instrument	~5 M cells per run	<40% of reads used weaker marker signal	>350,000 input cells
Scale QuantumScale	fixed cells or nuclei no instrument	~4 M cells per run	weaker marker signal	>350,000 input cells

Numbers: nine-platform head-to-head (Genentech), bioRxiv 2025.05.19.654974. Axis 4 — what rides along (immune receptors, protein, chromatin) — is kit-specific: read the kit sheet.
Ownership: 10x acquired Scale (Aug 2025); Illumina acquired Fluent (Jul 2024).

Four platforms against the four questions (nine-platform benchmark, bioRxiv 2025.05.19.654974)

Throughput costs doublets — and you pay in QC

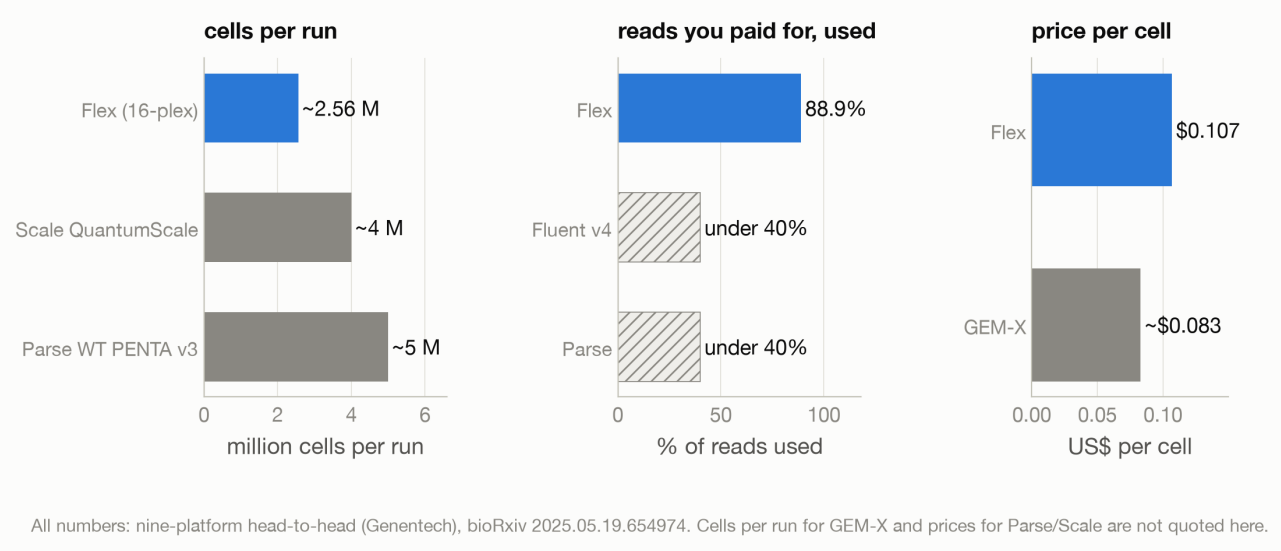
- GEM-X (10x): most cells recovered (71.1%), 24–30% more genes/UMIs per cell
- ~$0.083 per cell — and the most doublets of the nine, 10–11.6%
- More cells per run, more droplets holding two: Poisson, not bad luck
- A doublet can pass for a new cell type — the most convincing false cluster



Left: Poisson loading arithmetic (schematic). Right: GEM-X in the nine-platform benchmark

Fixed cells, millions per run: Flex, Parse, Scale

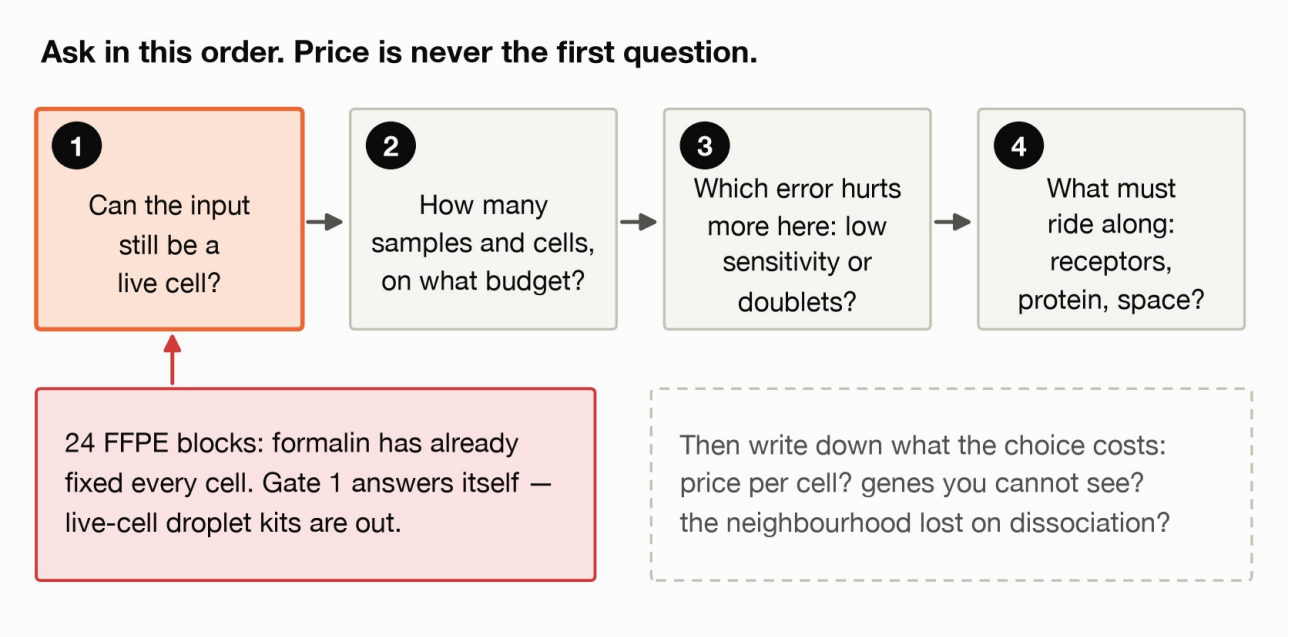
- Flex (10x): probes, FFPE-compatible, bankable, 16-plex, ~2.56 M cells/run
- Flex: 88.9% of reads used (Parse, Fluent v4 <40%), but $0.107/cell
- Parse (~5 M/run), Scale (~4 M/run): no instrument
- The catch: >350,000 input cells, and a weaker marker signal
- Probes see only what they target: nothing off the panel



Cells per run, reads used, price per cell (nine-platform benchmark, bioRxiv 2025.05.19.654974)

Exam question: twenty-four FFPE tumour blocks

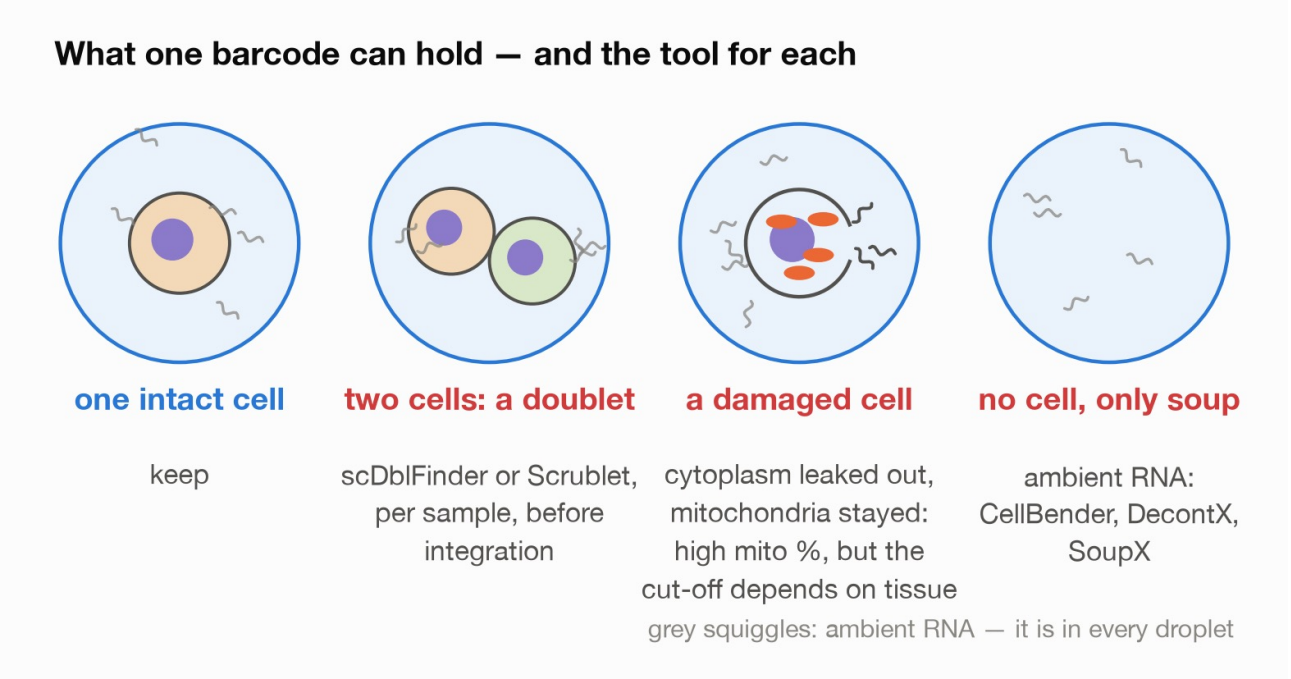
- Twenty-four FFPE (formalinle sabitlenmiş, parafine gömülü) tumour blocks; a fixed budget
- The research question: tumour–immune interaction
- Which platform, and what does the choice cost you?
- A language model will answer fluently and plausibly. Check its first assumption.



The order of the questions is the skeleton of the answer

QC thresholds are decisions about a tissue

- "Mito < 20%" is no rule: kidney and heart run high; single nuclei run near zero
- Ambient RNA (ortam RNA'sı): CellBender, DecontX, SoupX
- Doublets: scDblFinder or Scrublet, per sample, before integration
- Record every threshold in GENOMES N: cells in, cells out
- Tools benchmarked: Frontiers in Bioinformatics, 2026



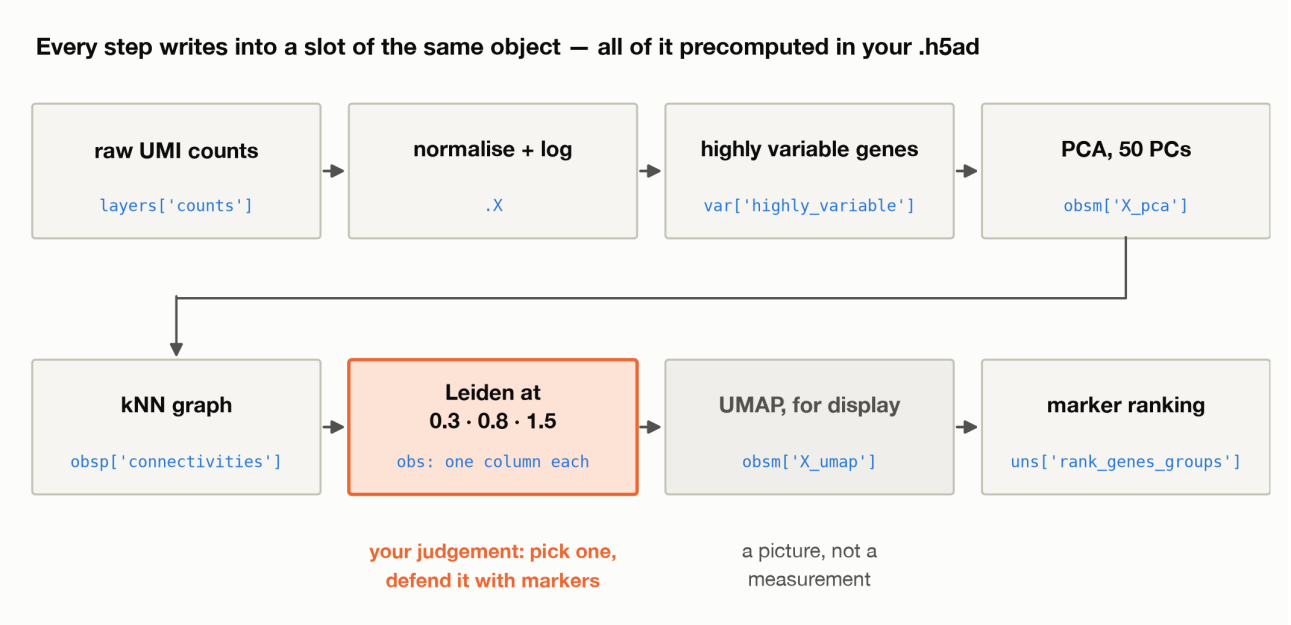
What one barcode can hold, and the tool for each (schematic)

The matrix: one AnnData object, six slots

- adata.X — cells × genes, normalised; rows are cells, the transpose of Week 7
- adata.layers['counts'] — the raw UMI counts, kept beside .X
- adata.obs — one row per cell (donor, condition…): adata.obs.head(3), Week 3
- adata.var — one row per gene: IDs, symbols, highly variable or not
- adata.obsm['X_pca'] (cells × 50) and adata.obsm['X_umap'] (cells × 2, display only)
- Saved as .h5ad; Seurat v5 is the same idea in R

The clusters: a resolution you choose and defend

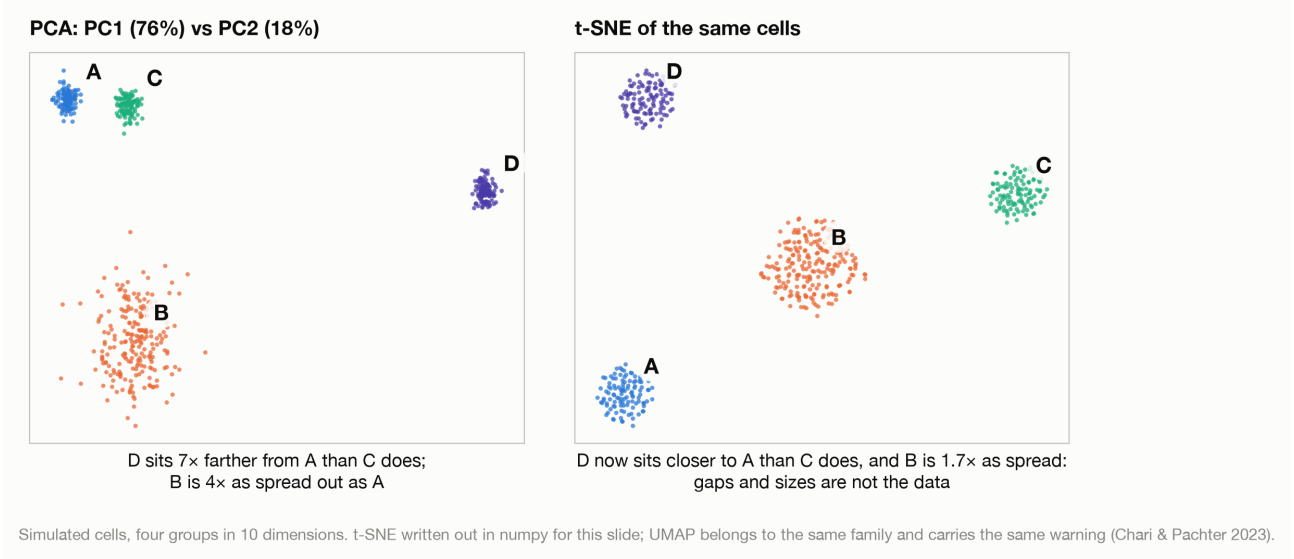
- PCA, then a k-nearest-neighbour graph over the PCs (Week 9)
- Leiden cuts the graph; resolution (çözünürlük) sets how finely
- Your file holds 0.3, 0.8 and 1.5 — no elbow, no "true" k
- Choose one; defend it with marker genes (belirteç genler)
- A cluster is a hypothesis about a cell type, not yet a cell type



Every step writes into a slot of the same object; the Leiden resolution is the judgement call

UMAP, second warning: a picture, not a measurement

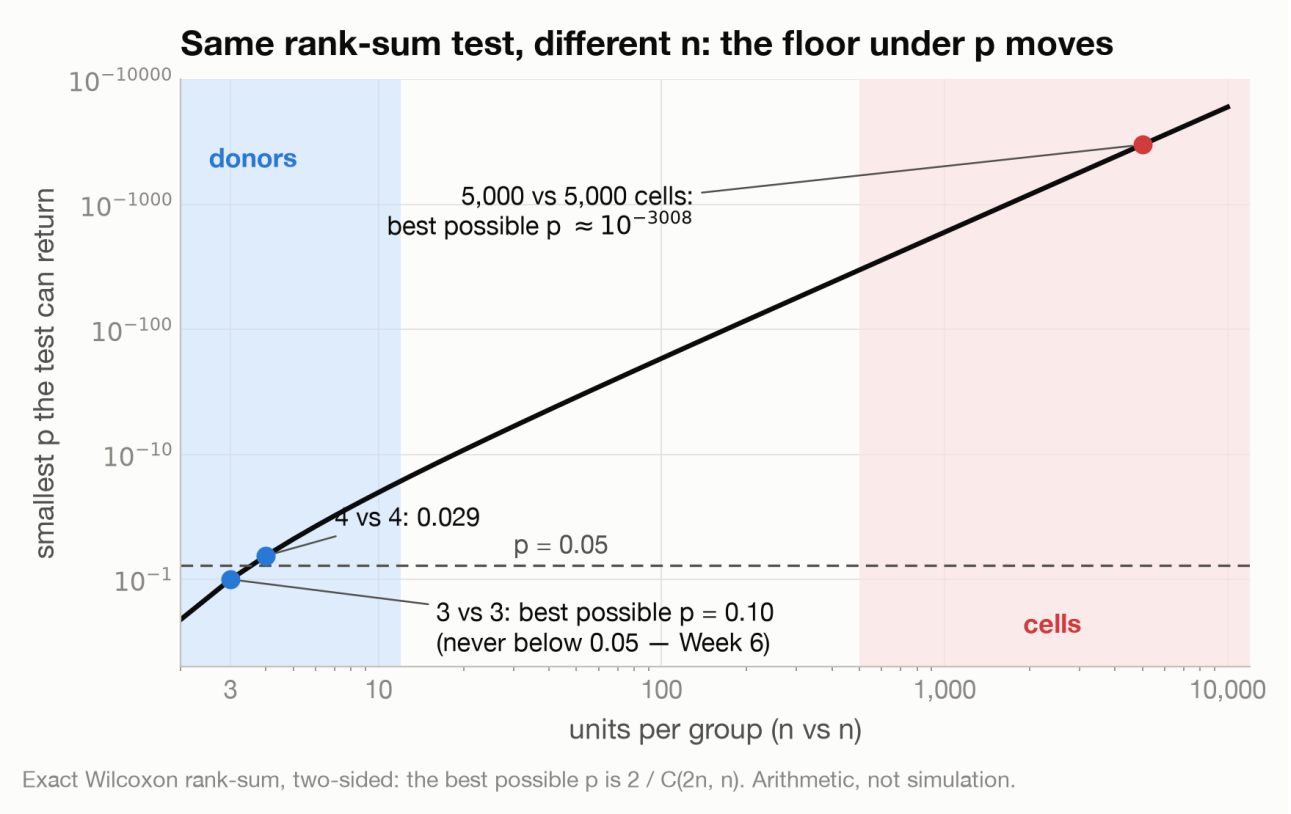
- UMAP draws the kNN graph in 2-D: neighbourhoods kept, geometry not
- Distances between clusters and cluster sizes mean nothing
- New seed or n_neighbors: new picture, same cells, same clusters
- Colour it by donor first: a one-donor island is not a cell type
- Chari & Pachter, PLoS Comput Biol 2023



Simulated cells: PCA keeps distances and spreads; t-SNE, UMAP's older cousin, does not

Revision: the rank-sum test, and the floor under p

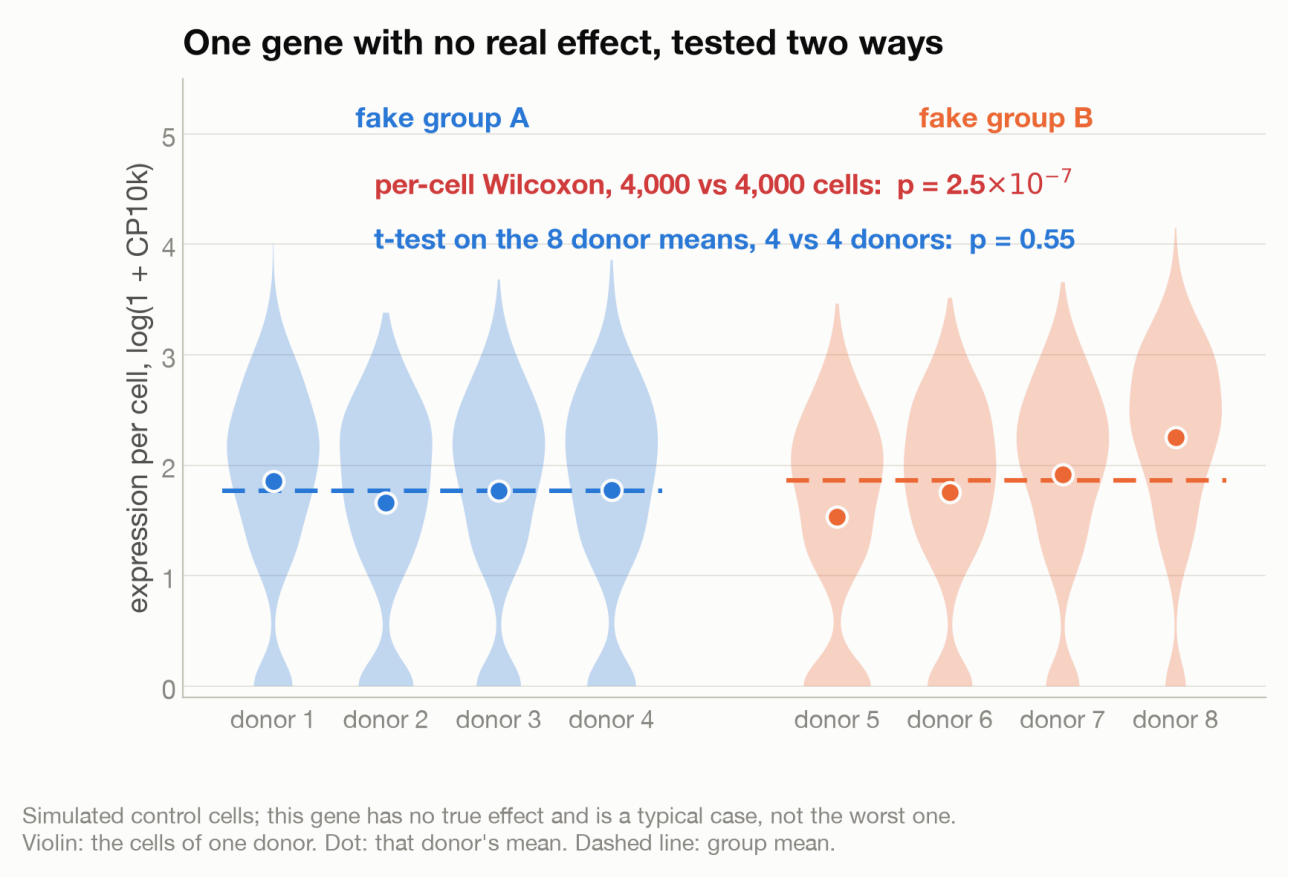
- Wilcoxon rank-sum (sıra toplamı testi): rank all values, sum one group's ranks
- Null: labels exchangeable — every split of the ranks equally likely (Week 6)
- 3 vs 3: 20 splits, best two-sided p = 0.10 — never below 0.05
- 5,000 vs 5,000 cells: best possible p ≈ 10^{-3008}
- The test is innocent. The unit is guilty.



The best p an exact rank-sum test can return depends only on n

Cells from one donor are pseudoreplicates

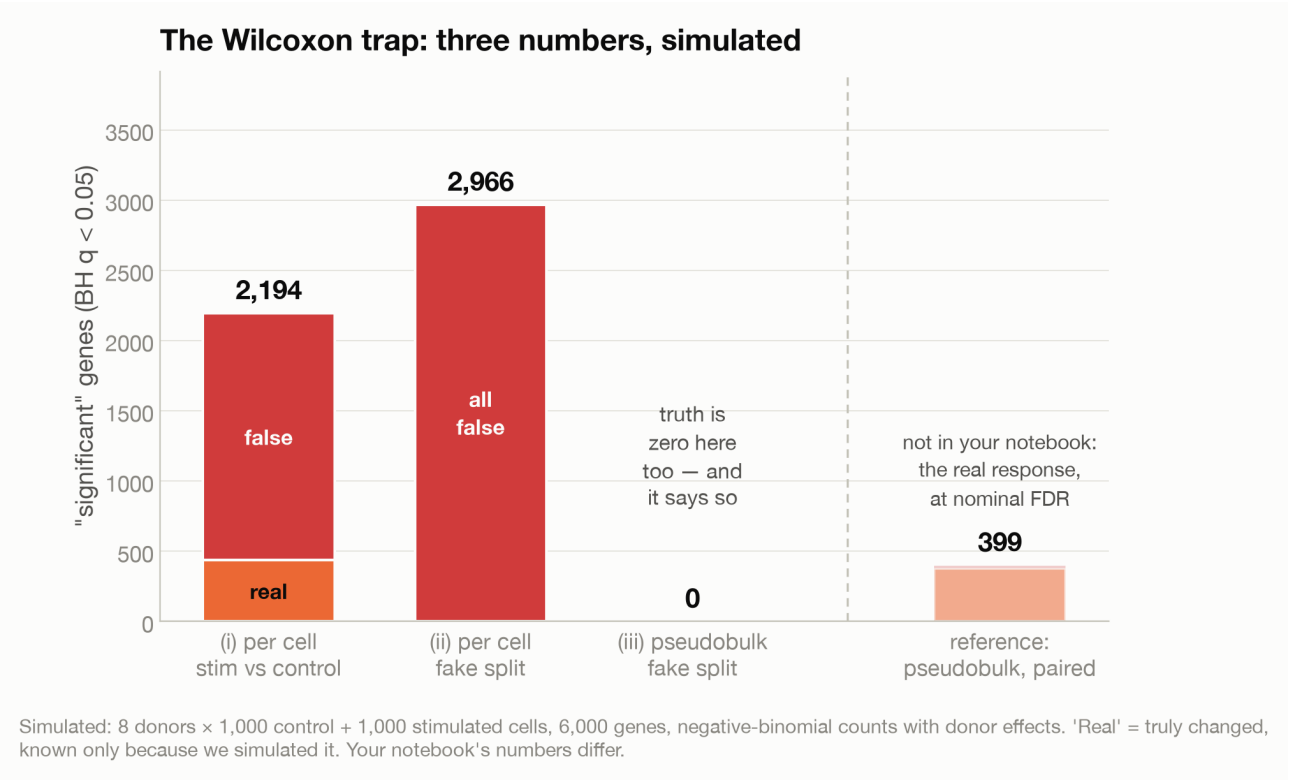
- Cells from one donor share its genotype, age, sampling day and batch
- 8,000 cells from 8 donors is n = 8 — Week 6, Case 2
- A per-cell test reads donor differences as your condition
- Squair et al. 2021: hundreds of false calls where nothing differs
- Pseudoreplication (sözde tekrarlama): the commonest single-cell error



One gene, no real effect: p = 2.5×10⁻⁷ across 8,000 cells, p = 0.55 across 8 donors (simulated)

Three numbers: the Wilcoxon trap, simulated

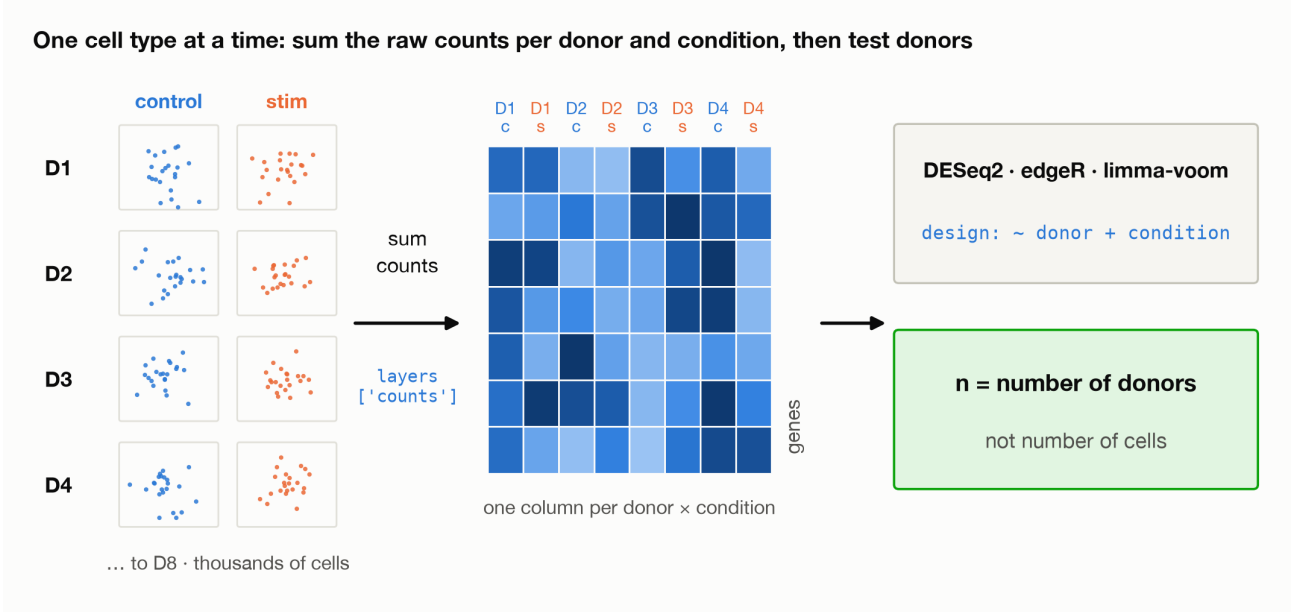
- (i) Per-cell Wilcoxon, stimulated vs control: thousands
- (ii) Same test, fake split of control donors, truth zero: still thousands
- (iii) Pseudobulk, same fake split, donor as n: almost none
- (ii) is Week 6's negative control: it convicts the method
- This afternoon: your own three, on real blood cells



Simulated: 8 donors × 2,000 cells, 6,000 genes — your notebook's numbers differ

Pseudobulk: aggregate to the donor, then use Week 7

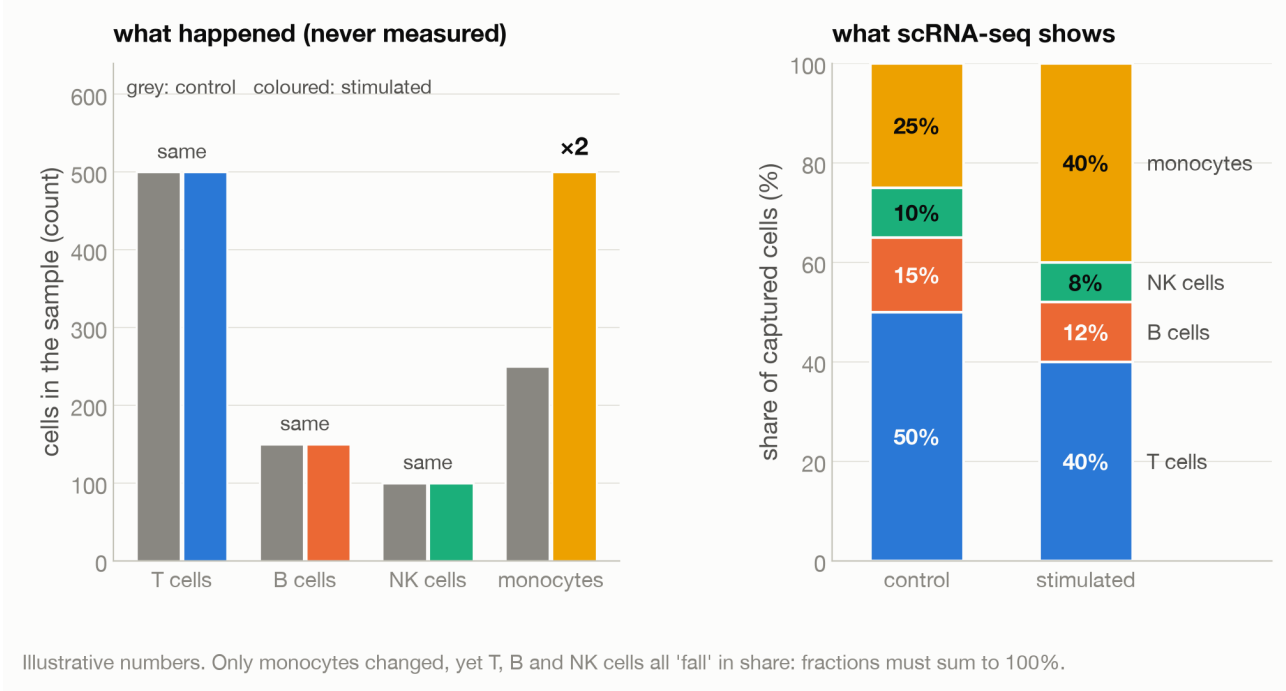
- Pseudobulk (sözde toplu): sum raw counts per sample × cell type
- DESeq2, edgeR or limma-voom; paired donors: ~ donor + condition
- One sample per condition: no valid test, however many cells
- rank_genes_groups for marker ranking is descriptive — fine
- Murphy & Skene 2022: pseudobulk ahead on a balanced measure



Cells → one profile per donor and condition → a Week 7 test with donors as n

The proportion illusion: fractions sum to one

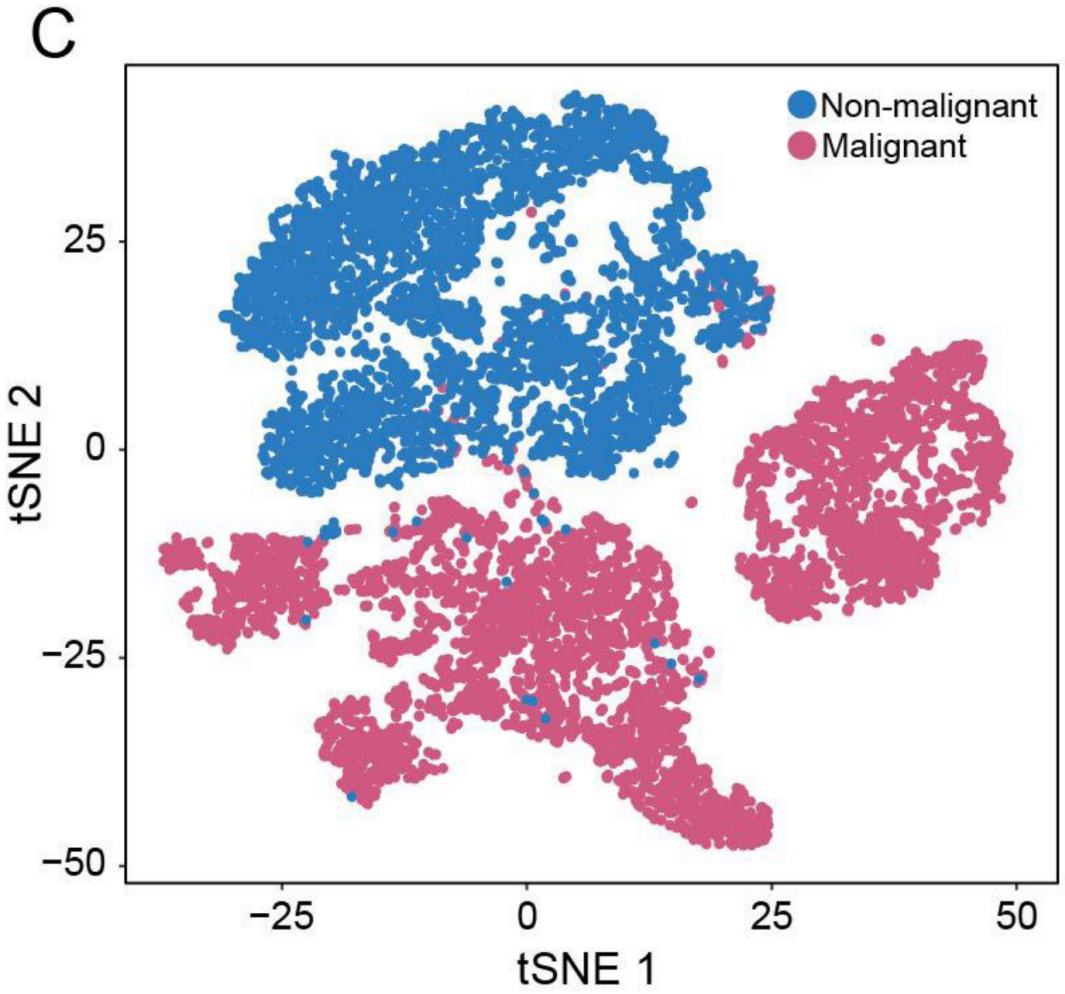
- Only monocytes change here: they double; T, B and NK do not
- As fractions, every population moves; three seem to shrink
- scRNA-seq shows shares only: loading, not tissue, sets cell numbers
- Per-type tests on shares report changes that never happened
- Named only: scCODA, propeller, Milo — samples as n



Illustrative numbers: one real change, four apparent ones

Why it matters: gastric cancer, cell by cell

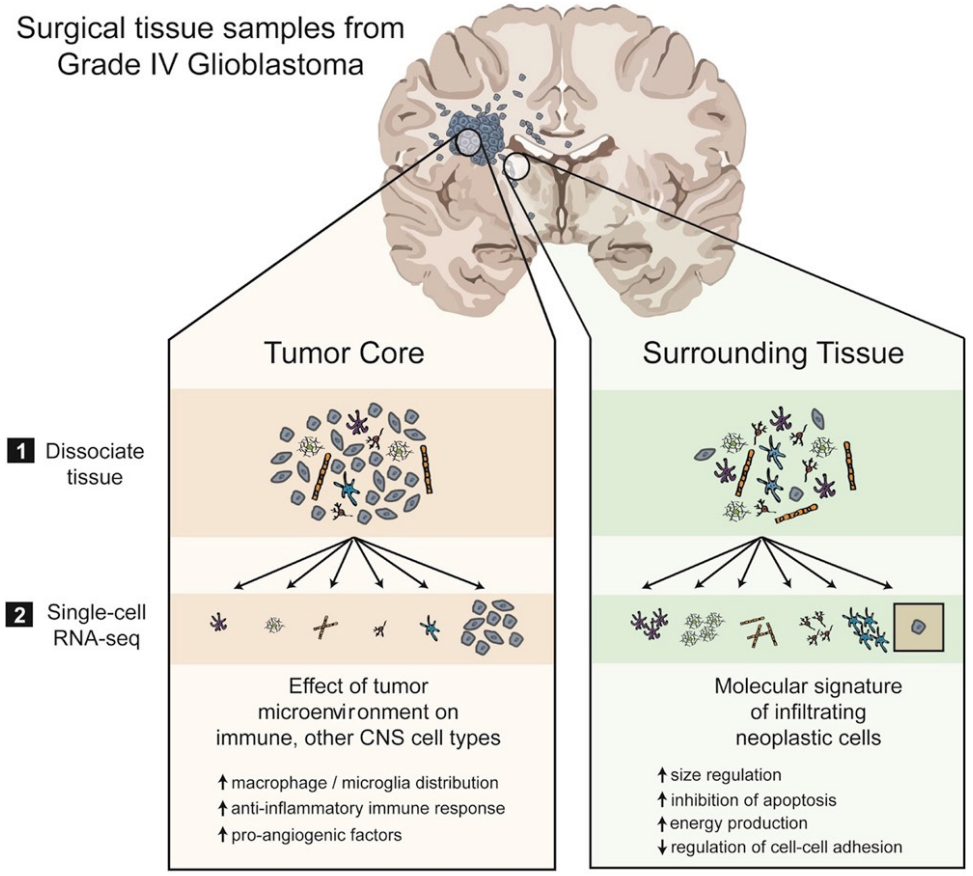
- Zhang et al. 2021: 9 gastric tumours and 3 non-tumour controls, 10x droplets
- Malignant and non-malignant epithelium separate; REG4, CLDN7 mark malignant cells
- Biomarkers resolved to one cell type; a bulk biopsy reports an average
- Before quoting a tumour-vs-normal list: count the patients behind it



Malignant (pink) and non-malignant (blue) epithelial cells, t-SNE (Zhang et al. 2021)

…and the cells at the edge of a glioblastoma

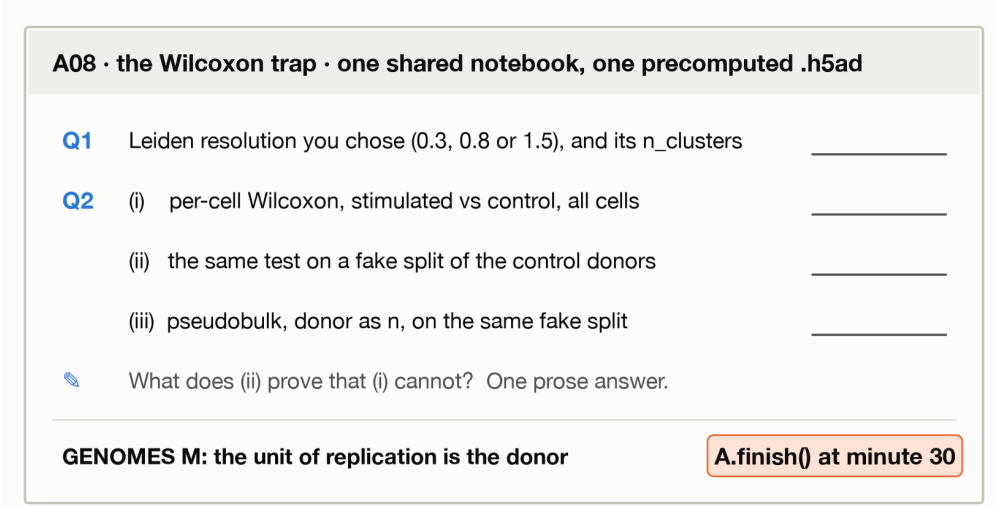
- Darmanis et al., Cell Rep 2017: glioblastoma core vs surrounding brain
- Neoplastic cells infiltrate the periphery, with a signature of their own
- Absent from a core biopsy; diluted in a bulk sample of the edge
- Same question before quoting the signature: how many patients?



Tumour core versus surrounding tissue in glioblastoma (Darmanis et al., Cell Rep 2017, graphical abstract)

This afternoon: A08, the Wilcoxon trap

- One notebook, one precomputed .h5ad: nothing installed, nothing clustered live
- Q1: pick a resolution (0.3 / 0.8 / 1.5), record n_clusters, defend it with markers
- Q2: three numbers — per-cell real, per-cell fake split, pseudobulk fake split
- One sentence: what does the fake number prove that the real one cannot?
- A.finish() at minute 30; GENOMES M: unit of replication = the donor
- Next week: no silhouette, no ARI — kNN accuracy and recall only



The receipt: Q1, three numbers, one sentence — A.finish() at minute 30